

● *Original Contribution*

DIAGNOSTIC PERFORMANCE OF CONTRAST-ENHANCED ULTRASOUND FOR OVARIAN CANCER: A META-ANALYSIS

YING WU, HONGLING PENG, and XIA ZHAO

Department of Gynecology and Obstetrics, West China Second Hospital, Sichuan University, Chengdu, People's Republic of China

(Received 26 September 2014; revised 8 November 2014; in final form 24 November 2014)

Abstract—This meta-analysis is the first study aimed at assessing the overall diagnostic performance of contrast-enhanced ultrasound for ovarian cancer. PubMed, Embase and Medline databases were systematically searched for relevant articles published up to June 2014. Data were pooled to yield summary sensitivity, specificity, diagnostic odds ratio and receiver operating characteristic curves using Meta-Disc Version 1.4 software. Ten independent studies with 579 ovarian tumors were enrolled in this meta-analysis. The pooled sensitivity, specificity and diagnostic odds ratio statistics were 0.89 (0.83–0.94), 0.91 (0.88–0.93) and 91.70 (41.41–203.05), respectively, and the area under the summary receiver operating characteristic curve was 0.9619 (standard error: 0.0125), all indicating that contrast-enhanced ultrasound has high diagnostic accuracy in differentiation of malignant from benign ovarian tumors. (E-mail: Wuyingxiong@163.com) © 2015 World Federation for Ultrasound in Medicine & Biology.

Key Words: Contrast-enhanced ultrasound, Diagnosis, Ovarian cancer, Meta-analysis.

INTRODUCTION

Ovarian cancer is the most lethal gynecologic malignancy (Siegel et al. 2012), partly because it is detected late, with greater than 70% of patients presenting at an advanced stage (Cohen et al. 2014). Five-year survival for all stages is 47%, and for advanced stages, <30% (Carter and Downs 2011). Early detection is one of the most important strategies for improving patient prognosis (Suh et al. 2012). In the early stage, ovarian cancer is usually asymptomatic, and moreover, only a small number of the relatively common ovarian masses detected by imaging techniques are malignant, which makes it crucial to differentiate benign from malignant ovarian masses. Transvaginal sonography usually is the initial diagnostic modality of choice for assessment of most adnexal masses; however, some masses, especially early-stage ovarian cancer, remain difficult to classify by conventional transvaginal sonography, even in experienced hands (Veyer et al. 2010; Wang et al. 2011; Xiang et al. 2013). Solid tumors and their metastases persist and

grow through angiogenesis, which is characterized by a neovascular network with irregularly branching vessels derived from pre-existing normal venules that contain numerous arteriolar–venous malformations without an intact basement membrane (Feldmann et al. 1999). Thus, imaging of vessels in tumors may help to assess the risk of ovarian cancer. However, radiologic assessment of tumor vascularity is not yet well established. The lack of screening tests for diagnosis of early-stage ovarian cancer is an important determinant of the mortality rate of this disease (Enakpene et al. 2009).

Contrast-enhanced ultrasound (CEUS), with the use of contrast agents consisting of gas microbubbles that are administered intravenously and remain intravascular, has been used to evaluate many tumors in the liver, kidneys, pancreas, breasts and other organs (Jakobsen et al. 2005), improving the characterization of tumor angiogenesis and perfusion. In addition, the kinetics of contrast agents in tumors can be evaluated objectively by quantifying time–intensity curve (TIC) parameters. A few studies have investigated the use of contrast-enhanced sonography in the differential diagnosis of malignant versus benign ovarian masses. Diagnostic accuracy in these published studies varied widely, with the sensitivity ranging from 74% to 100% and the specificity ranging from 42% to 98% (D'Arcy et al. 2004; Fleischer et al. 2009;

Address correspondence to: Ying Wu, Department of Gynecology and Obstetrics, West China Second Hospital, Sichuan University, Chengdu 610041, People's Republic of China. E-mail: Wuyingxiong@163.com

Huchon et al. 2012; Kupesic and Kurjak 2000; Marret et al. 2004; Orden et al. 2003; Testa et al. 2007, 2009; Veyer et al. 2010; Xiang et al. 2013), probably as a result of advances in technology, the heterogeneity of patient populations and so on. The overall accuracy of CEUS in the diagnosis of ovarian cancer has never been systematically assessed. The purpose of this study was to perform a systematic review and meta-analysis of published information to assess the overall diagnostic performance of CEUS in ovarian cancer.

METHODS

Literature search

PubMed, Embase and Medline databases were systematically searched for relevant articles on the diagnosis of ovarian masses using CEUS up to June 2014. Key words and medical subject headings were designed as follows: ovarian or adnexal; neoplasm or carcinoma or tumor or cancer or mass or lesion; contrast enhanced or contrast media or contrast agent; and ultrasound or ultrasonography or sonography.

Selection criteria

Inclusion criteria were as follows. (i) The study must relate to CEUS in differentiating between benign and malignant ovarian masses. (ii) The study must have been performed on human subjects. (iii) The study must be in the English language. (iv) The reference standard must be histopathologic findings or clinical diagnosis. (v) Sufficient data must be available in the fourfold (2×2) tables, corresponding to true positive (TP), true negative (TN), false positive (FP) and false negative (FN). (vi) The data or subsets of the data must not have been published more than once. (vii) The study population must be >10 patients.

Reviews, editorials, letters and case reports were excluded because of the limited data provided. When there were two articles by the same author(s) or from the same medical center with the same date, the article with the larger sample size was selected. Studies in which data were incomplete were excluded. Study selection was conducted by two researchers independently (Y.W. and H.P.). Disagreement was resolved by a third researcher (X.Z.) through discussion.

Extraction of data

The data recorded for each article included First author, publication year, country of study, mean age of patients, sonographic diagnostic criteria, contrast agent, number of lesions and TP, TN, FP and FN results. Data were extracted by two researchers independently (Y.W. and H.P.). Disagreement was resolved by a third researcher (X.Z.) through discussion.

Quality assessment

Methodologic quality was assessed independently by two researchers (Y.W. and H.P.) using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS). QUADAS criteria include 14 assessment items for systematic reviews of diagnostic accuracy studies. Each of 14 items was scored yes (score 1), no (score -1) or unclear (score 0). Study quality was defined as high when the QUADAS score was ≥ 11 (Whiting et al. 2003). Any disagreement between the two researchers was resolved by a third researcher (X.Z.) through discussion.

Statistical analysis

Between-study heterogeneity was estimated with Cochran's Q statistic and the inconsistency index (I^2) (Higgins et al. 2003). For a significant Q -test with $p < 0.1$ (Higgins and Thompson 2002) or $I^2 > 50\%$, considered as significant heterogeneity, the random-effect model was used; otherwise, the fixed-effect model was used. Diagnostic threshold effect was evaluated with receiver operating characteristic (ROC) space and the Spearman correlation coefficient between sensitivity and specificity. Representation of a typical "shoulder arm" pattern in a ROC space and a strong positive correlation between the log of sensitivity and log of $1 - \text{specificity}$ would suggest the presence of a threshold effect (Hu et al. 2014).

The pooled statistics of sensitivity, specificity, positive likelihood ratio (LR+), negative likelihood ratio (LR-) and diagnostic odds ratio (DOR) were calculated with corresponding 95% confidence intervals (CIs) in a fixed-effect model or random-effect model, as were their forest plots were. Summary ROC (SROC) curves with the area under the curve (AUC) and the Q^* index (the point on the SROC curve where sensitivity and specificity are equal) were obtained to summarize the overall diagnostic performance of CEUS in ovarian cancer. AUC values

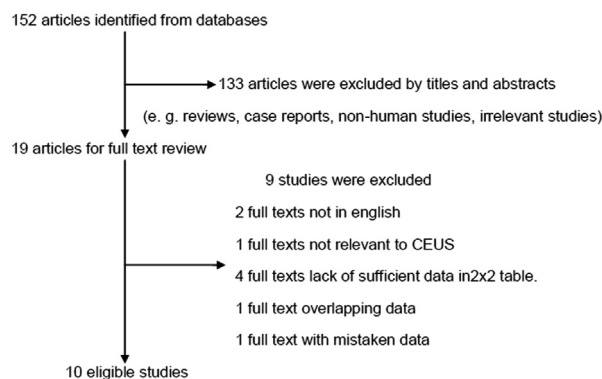


Fig. 1. Flowchart of literature search and study selection. Ten studies were included in this meta-analysis. CEUS = contrast-enhanced ultrasound.

Table 1. Characteristics of studies included in meta-analysis

Author(s)	Year	Country	Age (y)	Sonographic diagnostic criterion	Contrast agent	Number of lesions	2 × 2 Table				QUADAS score
							TP	FP	FN	TN	
Xiang <i>et al.</i>	2013	China	43.2 ± 12.6*	3-D CEUS score	SonoVue	51	8	1	0	42	11
Huchon <i>et al.</i>	2012	France	45.8 ± 15.1	3-D CEUS score	Levovist	99	9	9	2	79	13
Veyer <i>et al.</i>	2010	France	Malignant: 50.6 Benign: 49.3	TIC	SonoVue	52	10	4	2	26	12
Fleischer <i>et al.</i>	2009	USA	48.3 ± 2.1	TIC	Definity	36	10	1	0	25	12
Testa <i>et al.</i>	2009	Italy	52	TIC	SonoVue	72	26	1	9	36	13
Testa <i>et al.</i>	2007	Italy	41	Vascularization within papilla	SonoVue	33	9	14	0	10	11
Marret <i>et al.</i>	2004	France	46.2	TIC	Levovist	101	22	2	1	76	12
D'Arcy <i>et al.</i>	2004	UK	49	Vascular morphologic score	Levovist	20	4	1	0	15	12
Orden <i>et al.</i>	2003	Finland	Pre: 39 Post: 66	TIC	Levovist	70	13	4	1	48	11
Kupesic and Kurjak	2000	Croatia	Pre: 34 Post: 62	Vascular morphology	Levovist	45	12	2	0	31	14

FN = false negative; FP = false positive; Pre = pre-menopausal; Post = post-menopausal; QUADAS = Quality Assessment Tool of Diagnostic Accuracy Studies; TIC = time–intensity curve; TN = true negative; TP = true positive.

* Mean ± standard deviation.

ranged from 0.5 to 1.0, with values close to 0.5 indicating a poor test and AUC values of 1.0 indicating a perfect test (Hanley and McNeil 1982). The Q^* index, like the AUC, indicates how closely a test approaches the desired performance of 100% sensitivity and specificity. The higher the Q^* value, the better is the diagnostic test performance (Liu *et al.* 2007).

If significant heterogeneity emerged, to explore potential sources of heterogeneity, meta-regression analysis was then performed depending on the type of contrast agent (first generation vs. second generation) and sonographic diagnostic criteria (TIC vs. others); a p -value <0.05 was considered to indicate statistical significance (Gong *et al.* 2012). Deeks' funnel plot was produced to assess publication bias, with a p -value <0.1 suggesting significance (Deeks *et al.* 2005). All statistical analyses were performed using Meta-Disc Version 1.4 (Universidad Complutense, Madrid, Spain) and STATA Version 12.0 (STATA, College Station, TX, USA) software.

RESULTS

Characteristics and quality assessment of included studies

Initially, use of the pre-defined search terms yielded 152 publications. Of these, 133 studies were excluded on the basis of article titles and abstracts. After review of full texts for detail, 9 articles were excluded. Ultimately, 10 studies (D'Arcy *et al.* 2004; Fleischer *et al.* 2009; Huchon *et al.* 2012; Kupesic and Kurjak 2000; Marret *et al.* 2004; Orden *et al.* 2003; Testa *et al.* 2007, 2009; Veyer *et al.* 2010; Xiang *et al.* 2013) fulfilling the inclusion and exclusion criteria were included in this meta-analysis (Fig. 1); their main characteristics are summarized in Table 1. All included studies did obtain

informed consent from each study participant, and each study was approved by an ethics committee or institutional review board. Overall, 570 patients with 579 ovarian lesions were enrolled, of which 116 (20%) had malignant, 424 (73%) had benign and 39 (7%) had borderline tumors. Publication year of the eligible studies ranged from 2000 to 2013. One study (10%) was performed in Asia, 8 (80%) studies in Europe and 1 (10%) study in America. Nine of the 10 studies were prospective and 1 study was retrospective. Two main sonographic diagnostic criteria were used in the eligible studies, TIC and others. Two generations of contrast agents were used: the first-generation Levovist (Schering, Berlin, Germany) and the second-generation SonoVue (Bracco,

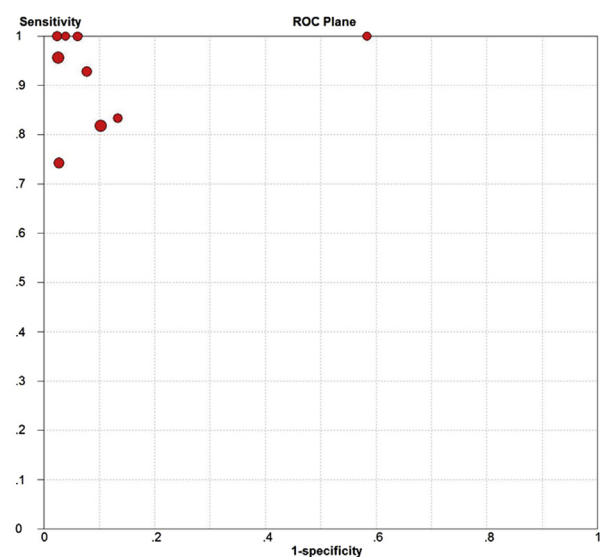


Fig. 2. Sensitivity versus 1 – specificity in receiver operating characteristic (ROC) plane for each eligible study.

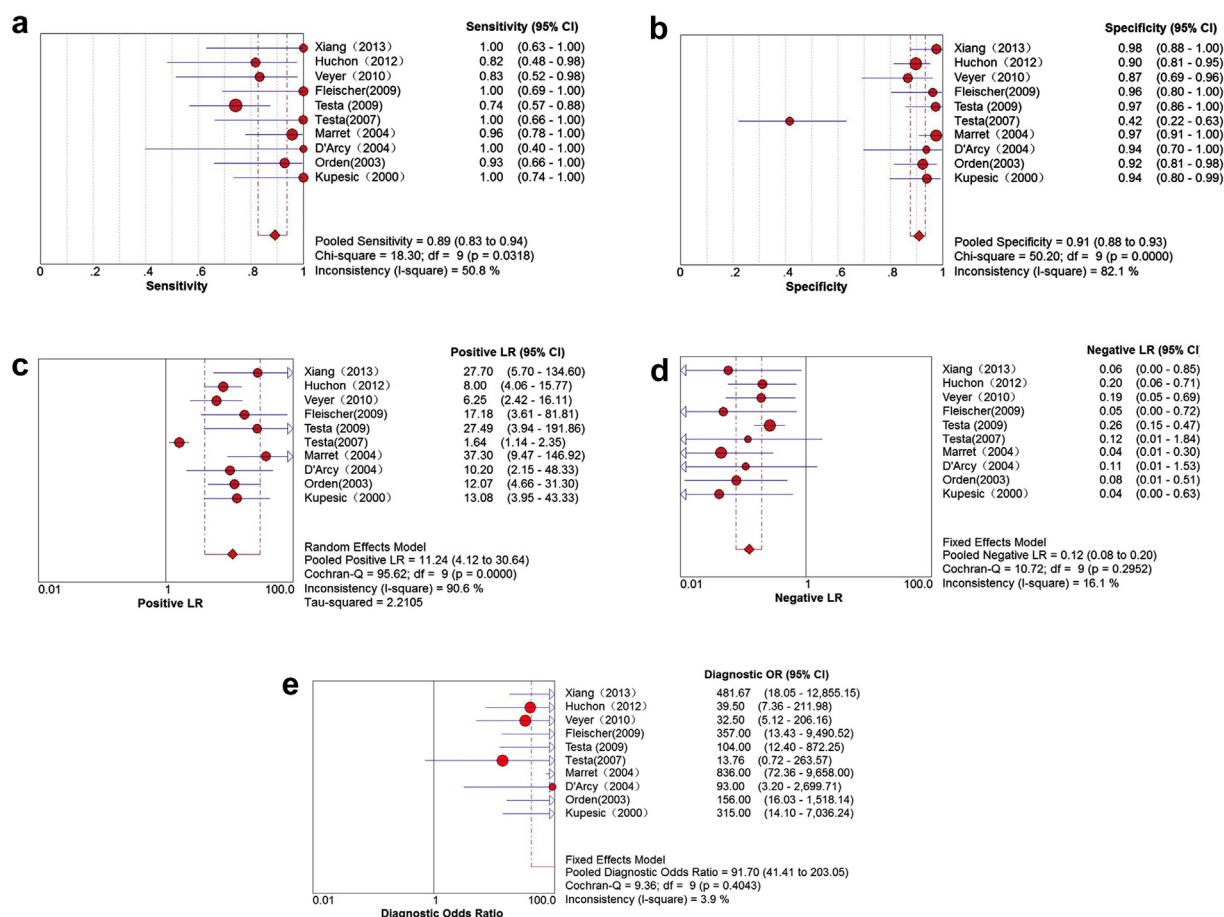


Fig. 3. Forest plots of the diagnostic accuracy of contrast-enhanced ultrasound. (a) Sensitivity. (b) Specificity. (c) Positive likelihood ratio. (d) Negative likelihood ratio. (e) Diagnostic odds ratio. CI = confidence interval; df = degrees of freedom; LR = likelihood ratio; OR = odds ratio.

Milan, Italy) and Definity (Lantheus Medical Imaging, North Billerica, MA, USA), accounting for 50% respectively. QUADAS scores of all included studies were ≥ 11 (range: 11–14), reflecting good methodologic quality, and are summarized in Table 1.

Quantitative data synthesis

Significant heterogeneity was observed across the studies in sensitivity (Q -test $p = 0.0318$, $I^2 = 50.8\%$), specificity (Q -test $p = 0.0000$, $I^2 = 82.1\%$) and LR+ (Q -test $p = 0.0000$, $I^2 = 90.6\%$), but not LR- (Q -test $p = 0.2952$, $I^2 = 16.1\%$) and DOR (Q -test $p = 0.4043$, $I^2 = 3.9\%$). There was no significant threshold effect in the present meta-analysis, as the ROC plane was not a typical “shoulder arm” pattern (Fig. 2) and the Spearman correlation coefficient was -0.248 with $p = 0.489$. Overall, the diagnostic accuracy of CEUS in differentiation of malignant from benign ovarian tumors was measured as the pooled sensitivity of 0.89 (0.83–0.94), pooled specificity of 0.91 (0.88–0.93), pooled LR+ of 11.24 (4.12–30.64), pooled LR- of 0.12 (0.08–0.20) and pooled

DOR of 91.70 (41.41–203.05). Forest plots of all these indices are provided in Figure 3. The results are plotted as a symmetric SROC curve, with an AUC of 0.9619 (standard error: 0.0125) and Q^* -value of 0.9072 (standard error: 0.0186), which indicate good diagnostic performance (Fig. 4). Meta-regression analysis including contrast agent type (first generation vs. second generation) and sonographic diagnostic criteria (TIC vs. others) was performed to investigate potential sources of heterogeneity. The results revealed that neither of the two characteristics (contrast agent type and diagnostic criteria) could explain potential sources of heterogeneity (Table 2). The Deeks funnel plot in Figure 5 reveals no evidence of significant publication bias with a p -value of 0.326.

DISCUSSION

To our knowledge, this systematic review/meta-analysis is the first aimed at assessing the overall performance of CEUS in diagnosing malignant ovarian tumors.

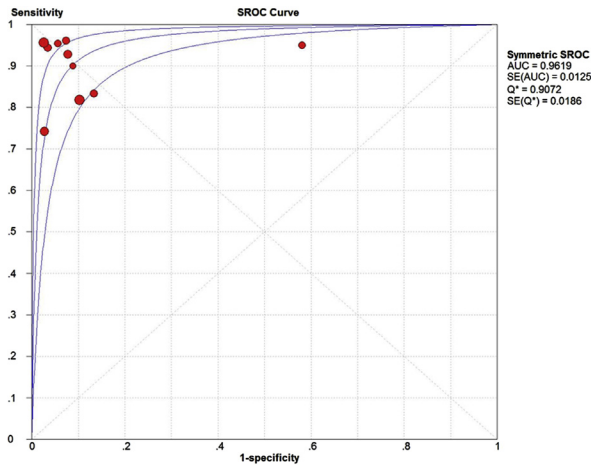


Fig. 4. Summary receiver operating characteristic curve for the diagnostic accuracy of contrast-enhanced ultrasound. SROC = summary receiver operator characteristic; AUC = area under curve; SE = standard error; Q^* = point at which sensitivity and specificity are equal.

In this meta-analysis, the 10 eligible independent studies comprised a total of 95 ovarian cancers, 387 benign ovarian tumors and 25 borderline ovarian tumors. The results revealed an overall sensitivity, specificity and DOR of 0.89 (0.83–0.94), 0.91 (0.88–0.93) and 91.70 (41.41–203.05), respectively, and the area under the SROC curve was 0.9619 (standard error: 0.0125), all indicating that CEUS is a valuable tool for differentiation of malignant from benign ovarian tumors with high diagnostic accuracy.

We tried to minimize any bias in this meta-analysis. The review was based on a thorough literature search and careful data extraction to avoid selection bias. QUADAS scoring was used to assess the methodologic quality of each eligible study, indicating that the included studies were all of good quality with data extraction scores ≥ 11 . Literature search, data extraction and quality assessment were conducted independently by two observers. The symmetric distribution of the included studies suggested there was no direct evidence of publication bias.

In recent years, CEUS has created new opportunities for diagnosis of ovarian cancer because it reveals tumor

neovascularity at the capillary level. CEUS is defined as traditional medical sonography using contrast agents consisting of gas microbubbles stabilized by a shell. It is a new technique for evaluating tumor vascularity and imaging tumor blood flow, with few adverse effects (Torzilli 2005); it is non-invasive, does not expose patients to radiation, is readily available and is cost-effective. Contrast agent injection improves the map of vascular anatomy and even allows detection of signals from blood vessels $<40 \mu\text{m}$ in diameter (Testa *et al.* 2009), thus effectively visualizing a larger number of vessels in malignant tumors than in benign tumors (Huchon *et al.* 2012). Additionally, the kinetics of contrast agents in tumors can be quantitatively assessed with TIC parameters. Compared with qualitative criteria highly dependent on experience, this examination is objective and reproducible and could be of value as a diagnostic aid to less experienced examiners (Veyer *et al.* 2010). However, quantification of the digitized vascular index has not been found to improve the pre-operative diagnosis compared with an examiner's subjective analysis (Gong *et al.* 2012; Veyer *et al.* 2010), and our meta-regression analysis also failed to reveal a relationship between the sonographic diagnostic criteria and the diagnostic odds ratio.

There are two generations of ultrasound contrast agents in clinical use: the first-generation Levovist for general use, and the second-generation SonoVue and Definity. In contrast to the first-generation agent, second-generation agents provide a substantial harmonic response at low acoustic pressure (Agati *et al.* 2001) without being destroyed, thereby providing continuous real-time vascular images of contrast enhancement for a prolonged time. Some studies have investigated how contrast-enhanced sonography discriminates malignant from benign ovarian masses through more prominent enhancement of malignant lesions (Huchon *et al.* 2012; Xiang *et al.* 2013). Borderline tumors are usually considered difficult to diagnose, probably more difficult than invasive and benign tumors. Contrast-enhanced sonography had higher performance in distinguishing invasive from non-invasive ovarian tumors, except for borderline ovarian tumors (Veyer *et al.* 2010). Therefore, additional studies are warranted to investigate further the characteristics of borderline ovarian tumors and improve the efficacy of CEUS in differentiating these tumors.

Recently, with technologic developments in ultrasound contrast agents and molecular biology, contrast-enhanced ultrasound with molecular targeted microbubbles (molecular ultrasound imaging) has been found to have great promise in the visualization and evaluation of tumor angiogenesis, both for the early detection of tumors (Mancini *et al.* 2013) and for monitoring early tumor response to anti-angiogenic therapy (Sorace *et al.* 2012) at the cellular and molecular levels, through

Table 2. Meta-regression analysis for potential sources of heterogeneity

Variance	Coefficient	SE	<i>p</i> -Value	RDOR	95% CI
Contrast agent	-0.644	0.8933	0.4944	0.53	0.06, 4.34
Sonographic diagnostic criterion	0.448	0.9759	0.6604	1.56	0.16, 15.72

CI = confidence interval; RDOR = relative diagnostic odds ratio; SE = standard error.

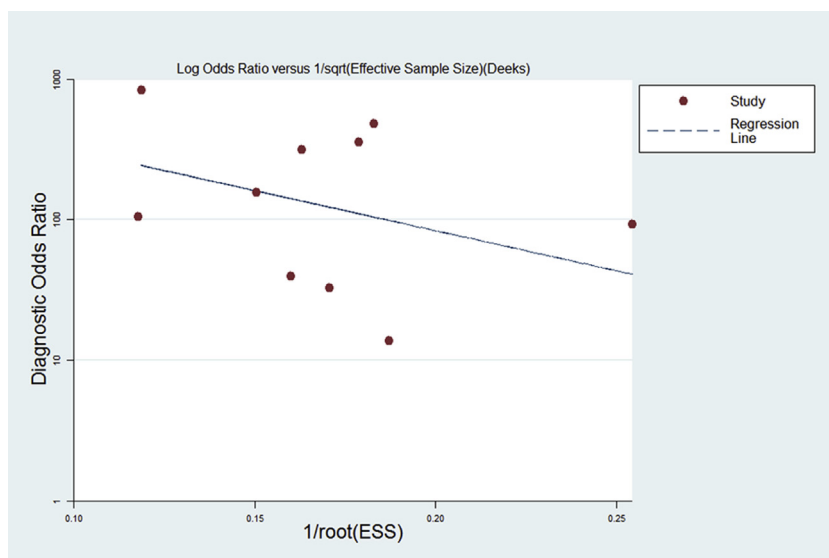


Fig. 5. Deeks' funnel plot of publication bias. ESS = effective sample size.

imaging of molecular targets expressed on the angiogenic vasculature of tumors. These microbubbles are retained in regions of disease by virtue of their shell composition/properties or by the conjugation of specific targeting ligands to their outer surface (Leong-Poi 2009). Different cancers and their endothelial cells overexpress certain receptor proteins. Microbubbles targeted to single or multiple receptors, such as $\alpha v \beta_3$ -integrin, vascular endothelial growth factor receptor 2 (VEGFR2) and endoglin, have been found to enhance visualization of angiogenesis through increased receptor binding in pre-clinical cancer models. In their study, Willmann et al. (2008) found that dual-targeted CEUS directed at both VEGFR2 and $\alpha v \beta_3$ -integrin improved *in vivo* visualization of tumor angiogenesis in a human ovarian cancer xenograft tumor model in mice. Having first been developed for molecular ultrasound imaging, targeted microbubbles are today used as gene or drug carriers in the process called ultrasound targeted microbubble destruction (UTMD), which has been reported to be a highly effective non-viral means of targeted gene transfer and drug delivery both *in vitro* and *in vivo* (Li et al. 2012; Ren et al. 2014). The research indicates that the mechanism underlying UTMD-mediated gene transfection or drug delivery is sonoporation (Chen et al. 2013). UTMD causes sonoporation, the formation of reversible nanometer or micrometer pores on cell membranes that increase the permeability of the membrane, effectively facilitating intracellular transfer of molecules, when the microbubbles are subjected to ultrasound irradiation. However, the biological and physical mechanisms underlying sonoporation are still not fully understood. Microbubbles loaded with anti-tumor genes or drug molecules target specific tissues and release

the therapeutic genes or drugs locally when exposed to ultrasound, increasing local accumulation of agents in tumor tissues and an enhanced anti-tumor effect with target specificity and high efficiency (Li et al. 2012). Furthermore, molecular targeted microbubbles have broad application prospects, not only for tumors, but also for inflammation, thrombi, heart diseases, central nervous system diseases and so on. Although these innovations are still in pre-clinical trials, we believe that targeted ultrasound contrast agents in molecular imaging and targeted therapy may further improve our ability to diagnose and treat human cancers in the future, to meet the growing clinical and research needs.

This meta-analysis had some limitations. First, there was notable heterogeneity between the individual studies. Heterogeneity can be due to random variation between studies, variation in characteristics and variation in the diagnostic threshold level required for a positive test result (Halligan et al. 2005). The results of the threshold effect test suggested there was no evidence of a threshold effect in the present study. Meta-regression analysis also did not detect potential sources of heterogeneity by adding covariates of contrast agent type and sonographic diagnostic criteria. Second, the relatively limited sample size in this meta-analysis may lead to insufficient statistical power to evaluate the diagnostic performance of CEUS in ovarian cancer. Third, failure to identify unpublished studies may affect the reliability and validity of our results. Finally, the lack of standardized imaging techniques, contrast agents and diagnostic criteria for CEUS may be why the individual studies included in this meta-analysis exhibited considerable variation in sensitivity and specificity.

CONCLUSIONS

This systematic review/meta-analysis suggests that CEUS has high diagnostic accuracy in the differentiation of malignant from benign ovarian tumors and may be used pre-operatively as a valid diagnostic method in patients suspected of having ovarian carcinoma. CEUS plays an important role in characterizing early-stage ovarian cancer and adnexal masses that are indeterminate at transvaginal sonography for more appropriate clinical management. Intravascular sonographic contrast agents improve the ability of sonography to identify microvessels, particularly early microvascular changes associated with early-stage ovarian cancer, leading to more complete delineation of the vascular anatomy and more precise definition of the biological behavior of an adnexal tumor. However, because of the limitations described above, large samples, multicenter studies and high-quality prospective trials are required to confirm our findings and further investigate this novel technique.

REFERENCES

- Agati L, Funaro S, Veneroso G, Volponi C, De Maio F, Madonna MP, Fedele F. Non-invasive assessment of myocardial perfusion by intravenous contrast echocardiography: State of the art. *Ital Heart J* 2001; 2:403–407.
- Carter JS, Downs LS Jr. Ovarian cancer tests and treatment. *Female Patient (Parsippany)* 2011;36:30–35.
- Chen ZY, Yang F, Lin Y, Zhang JS, Qiu RX, Jiang L, Zhou XX, Yu JX. New development and application of ultrasound targeted microbubble destruction in gene therapy and drug delivery. *Curr Gene Ther* 2013;13:250–274.
- Cohen JG, White M, Cruz A, Farias-Eisner R. In 2014, can we do better than CA125 in the early detection of ovarian cancer? *World J Biol Chem* 2014;5:286–300.
- D'Arcy TJ, Jayaram V, Lynch M, Soutter WP, Cosgrove DO, Harvey CJ, Patel N. Ovarian cancer detected non-invasively by contrast-enhanced power Doppler ultrasound. *BJOG* 2004;111: 619–622.
- Deeks JJ, Macaskill P, Irwig L. The performance of tests of publication bias and other sample size effects in systematic reviews of diagnostic test accuracy was assessed. *J Clin Epidemiol* 2005;58:882–893.
- Enakpene CA, Omigbodun AO, Goecke TW, Odukogbe AT, Beckmann MW. Preoperative evaluation and triage of women with suspicious adnexal masses using risk of malignancy index. *J Obstet Gynaecol Res* 2009;35:131–138.
- Feldmann HJ, Molls M, Vaupel P. Blood flow and oxygenation status of human tumors: Clinical investigations. *Strahlenther Onkol* 1999; 175:1–9.
- Fleischer AC, Lyschik A, Jones HW 3rd, Crispens MA, Andreotti RF, Williams PK, Fishman DA. Diagnostic parameters to differentiate benign from malignant ovarian masses with contrast-enhanced transvaginal sonography. *J Ultrasound Med* 2009;28:1273–1280.
- Gong TT, Hu DM, Zhu Q. Contrast-enhanced EUS for differential diagnosis of pancreatic mass lesions: A meta-analysis. *Gastrointest Endosc* 2012;76:301–309.
- Halligan S, Altman DG, Taylor SA, Mallett S, Deeks JJ, Bartram CI, Atkin W. CT colonography in the detection of colorectal polyps and cancer: Systematic review, meta-analysis, and proposed minimum data set for study level reporting. *Radiology* 2005;237: 893–904.
- Hanley JA, McNeil BJ. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology* 1982;143:29–36.
- Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21:1539–1558.
- Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;327:557–560.
- Hu Q, Wang XY, Zhu SY, Kang LK, Xiao YJ, Zheng HY. Meta-analysis of contrast-enhanced ultrasound for the differentiation of benign and malignant breast lesions. *Acta Radiol* 2015;56: 25–33.
- Huchon C, Metzger U, Bats AS, Bensaid C, Chatellier G, Azizi M, Lefrere-Belda MA, Dujardin A, Bernard JP, Lecuru F. Value of three-dimensional contrast-enhanced power Doppler ultrasound for characterizing adnexal masses. *J Obstet Gynaecol Res* 2012; 38:832–840.
- Jakobsen JA, Oyen R, Thomsen HS, Morcos SK. Safety of ultrasound contrast agents. *Eur Radiol* 2005;15:941–945.
- Kupesic S, Kurjak A. Contrast-enhanced, three-dimensional power Doppler sonography for differentiation of adnexal masses. *Obstet Gynecol* 2000;96:452–458.
- Leong-Poi H. Molecular imaging using contrast-enhanced ultrasound: Evaluation of angiogenesis and cell therapy. *Cardiovasc Res* 2009; 84:190–200.
- Li P, Zheng Y, Ran H, Tan J, Lin Y, Zhang Q, Ren J, Wang Z. Ultrasound triggered drug release from 10-hydroxycamptothecin-loaded phospholipid microbubbles for targeted tumor therapy in mice. *J Control Release* 2012;162:349–354.
- Liu J, Xu Y, Wang J. Ultrasonography, computed tomography and magnetic resonance imaging for diagnosis of ovarian carcinoma. *Eur J Radiol* 2007;62:328–334.
- Mancini M, Greco A, Salvatore G, Liuzzi R, Di Maro G, Vergara E, Chiappetta G, Pasquinelli R, Brunetti A, Salvatore M. Imaging of thyroid tumor angiogenesis with microbubbles targeted to vascular endothelial growth factor receptor type 2 in mice. *BMC Med Imaging* 2013;13:31.
- Marret H, Sauguet S, Giraudeau B, Brewer M, Ranger-Moore J, Body G, Tranquart F. Contrast-enhanced sonography helps in discrimination of benign from malignant adnexal masses. *J Ultrasound Med* 2004; 23:1629–1639. quiz 1641–1642.
- Orden MR, Jurvelin JS, Kirkinen PP. Kinetics of a US contrast agent in benign and malignant adnexal tumors. *Radiology* 2003;226: 405–410.
- Ren J, Zhang P, Tian J, Zhou Z, Liu X, Wang D, Wang Z. A targeted ultrasound contrast agent carrying gene and cell-penetrating peptide: Preparation and gene transfection in vitro. *Colloids Surf B Biointerfaces* 2014;121:362–370.
- Siegel R, Naishadham D, Jemal A. Cancer statistics, 2012. *CA Cancer J Clin* 2012;62:10–29.
- Sorace AG, Saini R, Mahoney M, Hoyt K. Molecular ultrasound imaging using a targeted contrast agent for assessing early tumor response to antiangiogenic therapy. *J Ultrasound Med* 2012;31: 1543–1550.
- Suh DH, Kim K, Kim JW. Major clinical research advances in gynecologic cancer in 2011. *J Gynecol Oncol* 2012;23:53–64.
- Testa AC, Timmerman D, Exacoustos C, Fruscella E, Van Holsbeke C, Bokor D, Arduini D, Scambia G, Ferrandina G. The role of CnTI-SonoVue in the diagnosis of ovarian masses with papillary projections: A preliminary study. *Ultrasound Obstet Gynecol* 2007;29: 512–516.
- Testa AC, Timmerman D, Van Belle V, Fruscella E, Van Holsbeke C, Savelli L, Ferrazzi E, Leone FP, Marret H, Tranquart F, Exacoustos C, Nazzaro G, Bokor D, Magri F, Van Huffel S, Ferrandina G, Valentin L. Intravenous contrast ultrasound examination using contrast-tuned imaging (CnTI) and the contrast medium SonoVue for discrimination between benign and malignant adnexal masses with solid components. *Ultrasound Obstet Gynecol* 2009;34: 699–710.
- Torzilli G. Adverse effects associated with SonoVue use. *Expert Opin Drug Saf* 2005;4:399–401.
- Veyer L, Marret H, Bleuzen A, Simon E, Body G, Tranquart F. Preoperative diagnosis of ovarian tumors using pelvic contrast-enhanced sonography. *J Ultrasound Med* 2010;29:1041–1049.
- Wang J, Lv F, Fei X, Cui Q, Wang L, Gao X, Yuan Z, Lin Q, Lv Y, Liu A. Study on the characteristics of contrast-enhanced ultrasound and its

- utility in assessing the microvessel density in ovarian tumors or tumor-like lesions. *Int J Biol Sci* 2011;7:600–606.
- Whiting P, Rutjes AW, Reitsma JB, Bossuyt PM, Kleijnen J. The development of QUADAS: A tool for the quality assessment of studies of diagnostic accuracy included in systematic reviews. *BMC Med Res Methodol* 2003;3:25.
- Willmann JK, Lutz AM, Paulmurugan R, Patel MR, Chu P, Rosenberg J, Gambhir SS. Dual-targeted contrast agent for US assessment of tumor angiogenesis in vivo. *Radiology* 2008;248:936–944.
- Xiang H, Huang R, Cheng J, Gulinaer S, Hu R, Feng Y, Liu H. Value of three-dimensional contrast-enhanced ultrasound in the diagnosis of small adnexal masses. *Ultrasound Med Biol* 2013;39:761–768.